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Assessing strategic use of artificial intelligence in self-regulated learning: Instrument development and evidence from Chinese university students

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Abstract

Despite the rapid integration of AI into students' self-regulated learning, rigorously validated instruments to assess their strategic AI use are critically lacking. Addressing this gap, the present study proposes a conceptual framework of self-regulated use of AI for learning, based on which a strategy inventory was developed and validated in two questionnaire surveys among Chinese university students (with valid $n=488$ and 707 respectively). Factor and Rasch analyses of the data identified eight factors underlying the instrument, with four pertaining to the regulation of AI-assisted task processing, and another four associated with the regulation of AI application. The corresponding subscales also demonstrated robust psychometric properties. Quantitative analyses further revealed imbalanced strategy usage. Specifically, output-focused metacognitive regulation was reported to be most frequent, whereas AI-supported self-assessment strategies were underutilized. Results also showed that there were gender and academic disciplinary differences in ethical and tool-focused metacognitive regulation; while degree of AI exposure was positively correlated with most strategy types, it had no significant correlation with ethical regulation. These findings highlight the need for differentiated interventions to address populational gaps in the AI-mediated learning process. This study contributes to a structured approach to understanding AI's role in self-regulated learning and calls for concerted measures to cultivate students' metacognitive and ethical awareness in AI application. Pedagogically, the validated instrument can be used by teachers to diagnose students' strategy use and inform targeted instruction. Students can also use it for self-assessment, enhancing their self-awareness and fostering a more balanced use of different AI strategies.

Keywords Artificial intelligence, AI strategy, Self-regulated learning, Scale development, Instrument validation, Higher education

Introduction

Self-regulated learning (SRL), the process in which learners proactively regulate their cognition, motivation, behaviors and emotions to attain learning goals, has long been proved as a notable contributor of learning achievement (e.g., Teng & Zhang, 2016; Zimmerman, 2002; Zimmerman & Schunk, 2011). Since the release of generative artificial intelligence (AI) tools such as ChatGPT and DeepSeek, students have been increasingly using them in their learning, substantially altering their SRL (e.g., Guo & Li, 2024; Jiang, 2024; Kim et al., 2024; Liu et al., 2024; Ou et al., 2024; Zou & Huang, 2024). Meanwhile, research evidence indicates that students vary in their ability to effectively use AI tools for learning, with some leveraging them for deep, meaningful engagement while others relying on them uncritically as shortcuts (Abdelhalim, 2024; Amaro et al., 2023; Tam, 2024; Yan, 2023; Yang et al., 2024). Therefore, it is of both pedagogical and research significance to quantitatively measure and analyze students' strategic uses of AI, so that patterns in such practices can be identified, potential causes pinpointed, and targeted guidance provided. However, most of the existing research investigating students' use of AI strategies in learning adopted a qualitative approach, and little has been done from a quantitative perspective; moreover, there is a lack of validated instruments for such quantitative research. This study aims to fill these gaps by 1) developing and validating such a self-report instrument and 2) analyzing the patterns in students' strategy usage.

The changing landscape of self-regulated learning and the need for new measuring instruments

Traditional self-regulated learning

Various models and frameworks have been developed in the past to describe SRL (e.g., Boekaerts, 1999; Greene & Azevedo, 2009; Pintrich, 2004; Winne, 2004; Zimmerman, 2002). Among them, Zimmerman's cyclic model and Pintrich's framework are the most well-known. The former conceptualizes SRL as a cyclic process involving three phases: *forethought*, *performance* and *self-reflection* (Zimmerman, 2002). Forethought phase involves key processes such as setting up learning goals, planning and selecting learning strategies, evaluating expected outcomes and judging self-efficacy in achieving those goals, while performance phase consists of the execution and monitoring of task or cognitive strategies (Zimmerman, 2002). Cognitive strategies are basic techniques used to process information in specific learning tasks (Poitras & Lajoie, 2013), such as searching, rehearsing, summarizing, elaborating, reasoning, organizing, transforming and integrating (Oxford, 2013; Pintrich, 2004; Winne, 2017). Finally, self-reflection phase encompasses processes such as self-evaluation of task performance or learning outcome and reaction to such reflection. Key processes in these three phases (e.g., planning, monitoring and self-evaluation) are also often collectively referred to as metacognitive strategies, which oversee and govern the deployment of cognitive strategies (Meijer et al., 2006; Pintrich, 2004).

Based on Zimmerman (2002), Pintrich (2004) further distinguishes monitoring from control within the performance phase, with the former referring to the ongoing metacognitive awareness of task, self and context, while the latter concerning the efforts put into controlling these aspects. Meanwhile, he differentiates between four areas of self-regulation: 1) *cognition regulation* involves key metacognitive and cognitive processes such as planning, monitoring, task processing, self-evaluation and causal attribution; 2)

motivation/affect regulation refers to activation of goal orientation, judgements of task value, difficulty and self-efficacy, and management of emotions during and after performing the task; *behavior regulation* involves allocation of effort and time to learning as well as seeking help from others (e.g., peers and teachers); 4) *context regulation* refers to managing or changing the learning environment to facilitate learning (Pintrich, 2004).

Different from the above two theories, Boekaerts' (1999) model posits three layers of self-regulation after taking into account the proximity of SRL to self and task. The central layer is *regulation of processing modes*, which refers to the choice of different cognitive strategies for task processing (Boekaerts, 1999) and is the most task-specific layer of regulation (Efklides, 2011). The intermediate layer is *regulation of the learning process*, which refers to employing metacognitive strategies to direct one's learning; the outermost layer is *regulation of self*, which refers to choice of goals and resources and represents the most general level of self-regulation that guides regulatory processes at the other two levels (Boekaerts, 1999; Efklides, 2011).

Adopting different perspectives, the above three theories complement each other in outlining important dimensions of SRL. In the past, numerous instruments have been developed to measure those dimensions of SRL (e.g., Pintrich, 1991; Teng & Zhang, 2016; Teng et al., 2022; Wang & Bai, 2017). However, these instruments cannot be directly applied to the current situation of AI-empowered SRL, given the substantial difference found between them, as discussed below.

AI-empowered self-regulated learning

The surge of generative AI tools such as ChatGPT in the past few years has significantly changed the landscape of students' SRL. With powerful capacity in processing and generating text, pictures and other types of information, these tools have been utilized by students in various manners to handle learning tasks, such as information searching (Abdelhalim, 2024; Adams et al., 2024), generating ideas or answers (Guo & Li, 2024; Jiang, 2024), and revising, grading and providing feedback on task performance (Tam, 2024; Zou & Huang, 2024).

Aside from these AI-assisted cognitive and metacognitive strategies used to tackle academic tasks, students also engaged in various monitoring, evaluative and adaptive processes of the AI tools they were using (Jiang, 2024; Liu et al., 2024; Ou et al., 2024; Xiao et al., 2025). A major impetus for such tool-oriented self-regulation relates to the drawbacks of those tools, which are known for providing superficial, biased and even fabricated information (Kohnke et al., 2023a; Liu & Xiao, 2025; Miao & Holmes, 2023). Such limitations necessitate students to critically assess and selectively adopt AI-generated content to achieve desirable outcomes (Bearman et al., 2024; Rad et al., 2024). Given that the self-regulated processes surrounding AI and its application (e.g., monitoring and evaluating AI output) substantially differ from those involving using it to address learning tasks (e.g., using AI tools to monitor and evaluate task performance), a distinction is made between them in the present study and a framework about self-regulated use of AI for learning is thus proposed (see Fig. 1).

Similar to Boekaerts' (1999) three-layer model, the proposed framework also postulates three levels of regulation for AI-mediated learning, varying according to their proximity to task and self. Different from Boekaerts (1999), however, our framework assumes that both AI-assisted cognitive and metacognitive strategies (e.g., Jiang, 2024;

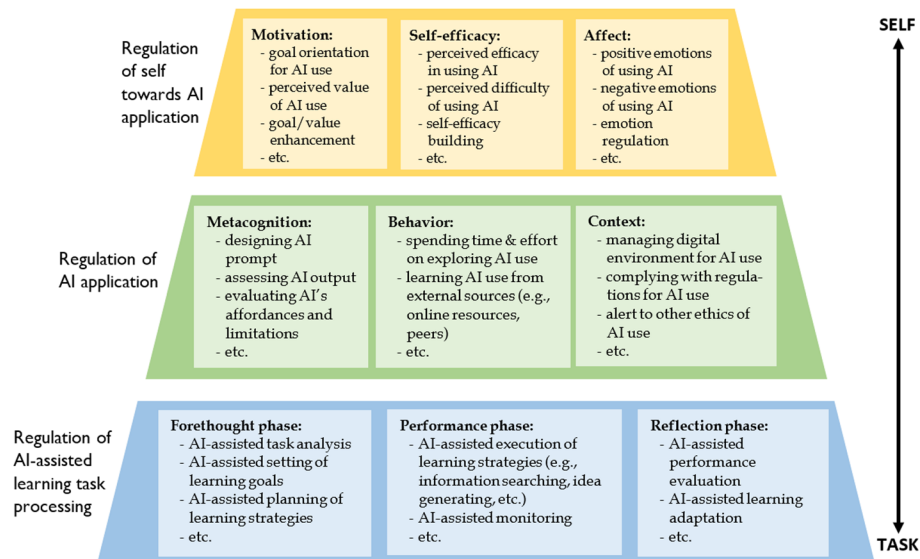


Fig. 1 A proposed framework of self-regulated use of AI for learning

Tam, 2024; Zou & Huang, 2024) conducted in the three phases of task completion (Zimmerman, 2002) constitute the bottom level of regulation, i.e., regulation of AI-assisted learning task processing. The intermediate level is the regulation of the AI application *per se* in the metacognitive, behavioral and contextual domains (Pintrich, 2004). Metacognitive regulation of AI application includes processes such as evaluating AI output and AI tools' strengths and weaknesses (e.g., Jiang, 2024; Liu et al., 2024; Ou et al., 2024); behavioral regulation refers to dedicating time and effort to learning AI application as well as engaging in social behaviors (e.g., help-seeking and peer learning) to facilitate such application (Pintrich, 1991, 2004; Xiao et al., 2025); contextual regulation involves structuring the environment for AI application and, most importantly, complying with the regulation, policies and ethics of using AI (Ng et al., 2023; Yan, 2023). The third level is the regulation of self, including motivation, self-efficacy and affect associated with AI utilization (Pintrich, 2004). In line with Boekaerts (1999), this is the most general level of regulation, as it determines the overall personal orientation, willpower and affection for using AI.

Need for new instruments to measure self-regulated use of AI for learning

Despite AI's growing presence in SRL, research evidence shows that students seemed to differ in their efficiency in leveraging those tools to learn, with some knowing how to retrieve useful information from them and employ them to engage in meaningful and deep learning, while others tending to adopt their output uncritically and utilize them as learning shortcuts (Abdelhalim, 2024; Amaro et al., 2023; Tam, 2024; Yan, 2023; Yang et al., 2024). Such difference has been attributed to different levels of self-regulation, including motivation (Tam, 2024), agency (Yang et al., 2024) and metacognitive awareness (Abdelhalim, 2024). Therefore, it is crucial for teachers, educational researchers and policy makers to further quantitatively investigate, using reliable instruments, how students engage with AI in conducting SRL and which practices are truly effective in facilitating SRL, so that targeted guidance can be provided.

A search of the literature revealed that, nevertheless, rigorously validated instruments that can be readily used to quantitatively survey AI strategy use are scant. For many survey studies, AI application was measured only at a holistic level, such as frequency and duration of usage (e.g., Bouzar et al., 2024; Elshaer et al., 2024; Stöhr et al., 2024). While some studies such as Abdelhalim (2024) and Von Garrel and Mayer (2023) employed items to survey students' specific uses of AI, they did not detail their development and validation process. In fact, most of the extant empirical studies examining students' use of AI strategies mainly adopted a qualitative approach, through methods such as reflective journals or learning logs (e.g., Jiang, 2024; Xiao et al., 2025; Zou & Huang, 2024), interviews (e.g., Mahapatra, 2024; Tai, 2024), written open-ended questions (e.g., Dizon, 2024; Essien et al., 2024), think-aloud (e.g., Liu et al., 2024), and screen recording or chatlogs (e.g., Guo & Li, 2024; Nguyen et al., 2024; Yan, 2023). A new, robust instrument that allows for quantitative data collection is thus strongly needed. To this end, we conducted a large-scale search of AI strategies reported in the literature. In the following sections we will review those strategies, focusing on those related to the bottom and middle levels of regulation shown in Fig. 1, whereas the top level of regulation is not covered in the present study given that there are already a number of instruments measuring its constituent components (e.g., Chan & Zhou, 2023; Wang et al., 2023; Xia et al., 2022; Yurt & Kasarci, 2024).

AI-assisted task processing strategies

Recent studies have identified a variety of AI strategies students adopt to assist their academic tasks. Among them, the majority seems to pertain to the performance phase of task completion, while those belonging to the forethought and reflection phase were less frequently reported.

Performance phase strategies

During the performance phase, students reported using AI tools to search for information (Abdelhalim, 2024; Kim et al., 2024; Su et al., 2023; Xiao et al., 2025), generating initial ideas or inspirations (Guo & Li, 2024; Jiang, 2024; Kim et al., 2024; Mahapatra, 2024; Nguyen et al., 2024), identifying potential topics (Abdelhalim, 2024), and inquiring the meaning of unfamiliar concepts involved in tasks (Abdelhalim, 2024; Kim et al., 2024; Zou & Huang, 2024). When studying those concepts, students reported asking AI tools to provide extensive examples, detailed explanations (Abdelhalim, 2024; Liu et al., 2024), or straightforward and comprehensible descriptions (Adams et al., 2024; Ou et al., 2024). For research-oriented tasks, students also employed AI tools to help summarize and synthesize relevant reading material (Essien et al., 2024; Yang et al., 2024), choose research methods (Xiao et al., 2025), refine research instruments (Abdelhalim, 2024), and generate references (Ou et al., 2024). For dissertation writing, some students were found to use AI tools to generate multiple outlines for comparison or to evaluate their own outline (Zou & Huang, 2024).

While the above strategies generally demonstrate students' process-oriented uses of AI, in other cases students were found to aim for more instrumental purposes. For example, Malaysian students in Adams et al. (2024) described how they used ChatGPT to elaborate on their points and offer them ideas, which helped them quickly complete their assignments without wasting time on thinking. Similarly, students in Kim et al

(2024) reported using ChatGPT to generate sentences for assignments based on initial drafts, while in Essien et al. (2024), some students employed AI tools to streamline their research process and speed their project completion progress. Lo et al. (2024) commented that such utilitarian strategies would be adopted when students feel lazy and prefer a shortcut to task completion.

AI tools are also frequently used by non-native-speaking students to address their linguistic deficiencies in handling tasks in the target language (e.g., English), by paraphrasing and summarizing the source text (Abdelhalim, 2024; Yan, 2023), or translating text between their mother tongue and the target language (Guo & Li, 2024; Ou et al., 2024). To improve the quality of their writing, students utilized various tools for language editing (Abdelhalim, 2024; Kim et al., 2024; Nguyen et al., 2024; Ou et al., 2024; Su et al., 2023; Tam, 2024). Such editing included detailed checking for issues with spelling, grammar (such as tense), word choice, language style, clarity, and cohesion (Mahapatra, 2024; Ou et al., 2024; Tam, 2024; Xiao et al., 2025), and direct polishing of the entire draft with one click (Abdelhalim, 2024).

Forethought and reflection phase strategies

Much less evidence exists in the literature regarding students' employment of strategies during these two phases. AI tools were reported being used by university students to plan their essays by breaking down the whole writing process into more specific stages during the forethought phase of task processing (Kim et al., 2024). Empirical studies also disclosed that during the reflection phase, students sought feedback from AI tools on both the language and content of their task products, such as word choice and sentence structures (Xiao et al., 2025), research questions (Abdelhalim, 2024), content and structure (Zou & Huang, 2024), and overall presentations of the essay (Jiang, 2024; Nguyen et al., 2024). Some asked ChatGPT to help them interpret assignment requirements as well as grade their essays based on those requirements (Abdelhalim, 2024; Tam, 2024).

AI application regulatory strategies

Aside from utilizing AI tools to help processing academic tasks, students were also found to be regulating their AI application process. Such regulatory strategies can be categorized into three groups, i.e., metacognitive, behavioral and contextual regulation, based on Pintrich's (2004) taxonomy.

Metacognitive regulation

Beyond using AI to facilitate their learning, students also recognized the need to metacognitively regulate their use of AI to achieve better outcomes. In empirical studies, students reported purposeful uses of strategies about prompt engineering to achieve more effective answers, such as providing sufficient context in prompts, using structured prompts progressing from general to specific instructions, offering examples, setting a clear focus or boundary for output, asking follow-up questions, as well as adjusting or refining prompts following output evaluation (Abdelhalim, 2024; Nguyen et al., 2024; Xiao et al., 2025; Yang et al., 2024). Based on such experiences, they reflected on the effectiveness of those prompting strategies and adapted their subsequent strategy use accordingly (Xiao et al., 2025).

Empirical evidence shows that upon receiving output, students asked AI tools for confirmation or elaboration (Abdelhalim, 2024; Liu et al., 2024), evaluated output's usefulness against personal knowledge, expectations or preference (Liu et al., 2024; Xiao et al., 2025), examined output's reliability through cross-referencing or scrutinizing the original sources (Ou et al., 2024; Xiao et al., 2025), and rejected, modified or selectively adopted the output (Jiang, 2024; Tam, 2024). For instance, almost all students in Liu et al. (2024)'s study noticed that sub-claims generated by ChatGPT lacked relevant examples and thus requested it to add examples. In Tam's (2024) study, students showed concerns over the authenticity of references provided by ChatGPT and cross-checked them through Google Scholar; furthermore, high achievers selectively adopted AI's feedback and took a more reflective approach in learning from those feedback (Tam, 2024).

Beyond specific AI output, students were also found to appraise different AI tools' strengths and weaknesses based on their use experiences (Jiang, 2024; Xiao et al., 2025), and such appraisals helped them update their metacognitive knowledge about AI application and adjust their AI strategies timely (Xiao et al., 2025). Constantly engaging in such evaluative processes also helps develop a critical mindset. For instance, students in Essien et al. (2024) believed the process of interacting with AI tools allowed them to develop their critical thinking skills by accessing different perspectives and identifying logical fallacies in AI-generated information.

Behavioral regulation

Another source of updating metacognitive AI knowledge is through proactive exploration. As an emerging technology, generative AI requires appropriate prompt engineering and tool selection for efficient utilization (Miao & Holmes, 2023; Tan et al., 2023). However, many teachers and students reported a lack of relevant skills and knowledge (Kim et al., 2024; Kohnke et al., 2023b). To bridge such gaps, students made different efforts, including playing with the different functions of AI tools at hand, learning from external sources such as social media, online videos, webpages, peers, as well as exploring new tools (Yan, 2023; Xiao et al., 2025). Such self-regulatory strategies are aligned with the behavioral regulation in Pintrich (2004), which includes aspects such as dedication and management of effort and time to learn, as well as learning or seeking help from others. The updated knowledge base in turn would affect subsequent AI use in SRL (Flavell, 1979; Xiao et al., 2025).

Contextual regulation

A salient feature of contextually regulating the use of AI for learning in the literature concerns the awareness of and compliance with AI ethics. An increasing number of studies have disclosed students' concerns about the fact that generative AI can be used for plagiarism, which is detrimental to academic integrity and educational fairness (Yan, 2023). Such concerns have prompted some students to use those tools in a more self-scrutinized or cautious manner (Zou & Huang, 2024). Some students also voiced that such tools should only be used as sources of help or reference instead of quick answers (Ou et al., 2024; Tossell et al., 2024), and that proper guidance and regulations be imposed to prevent unethical use (Yan, 2023; Zou & Huang, 2024). In short, students' ethical awareness regarding AI may influence their SRL involving it, and such ethical regulation represents their active interaction with the social context for AI application.

In recent years, studies such as Jang et al. (2022) and Wang et al. (2025) have developed and validated scales to measure university students' AI ethics. However, these scales are targeting more at students' overall ethical awareness in AI's general usage (e.g., AI for social good), rather than the strategies they take to regulate their learning process out of ethical considerations.

Group and individual differences in students' AI experience

Beside the well-documented uses of various AI strategies, recent survey studies also disclosed significant disparities in students' experience with AI across genders, academic disciplines and individuals globally.

Gender differences

Regarding gender, it was found that male students exhibited higher frequencies of AI tool usage, broader application scopes and more favorable perceptions of AI's role in learning compared to their female counterparts (Møgelvang et al., 2024; Stöhr et al., 2024). They also expressed greater interest in AI technologies and stronger beliefs about their relevance to future careers (Møgelvang et al., 2024). Conversely, female students' AI usage was less frequent and predominantly limited to text-related tasks; they also voiced heightened concerns about over-reliance, privacy risks and adverse impacts on critical and independent thinking, and expressed a stronger need for guidance on appropriate usage (Bouzar et al., 2024; Møgelvang et al., 2024; Stöhr et al., 2024). Within the specific area of AI ethics, females students demonstrated elevated ethical awareness (Wang et al., 2025) and stronger senses of fairness, privacy and non-maleficence in AI utilization for general purposes (Jang et al., 2022).

Disciplinary and individual differences

University students' AI experience was also found to vary across academic disciplines. Technology and engineering students reported significantly higher AI usage frequencies than peers in humanities and medicine (Stöhr et al., 2024). Social science students, compared to those in natural sciences, exhibit greater susceptibility to peer influence in adopting AI tools (Elshaer et al., 2024). For students' ethical reflective practice, while no difference was found across disciplines, individual difference emerged, as frequent AI users were found to engage in more ethical reflection than infrequent ones (Wang et al., 2025). Additionally, increased AI exposure was found to correlate positively with students' perceptions of AI's facilitative role in education (Stöhr et al., 2024).

Collectively, prior research underscores how individual practices, educational backgrounds, and demographic factors contribute to heterogeneous AI experiences among students. Investigating these disparities is imperative for equitable AI integration in education, as such insights enable targeted interventions tailored to group and individual characteristics (Stöhr et al., 2024; Wang et al., 2025). However, existing studies have yet to explore differences in students' deployment of specific AI strategies in learning, representing another critical gap in the literature.

Methodology

Instrumentation

From the abovementioned literature, we initially collected 115 strategies of using AI for learning. They were then classified according to the abovementioned two main categories, with 71 strategies related to addressing pedagogical tasks and 44 strategies about regulation of AI application. Within each category, we further scrutinized each item for potential content overlap. Through iterative examination and discussions, we deleted redundant items and merged similar ones to avoid responding fatigue and modified the wording of some other items for better clarity, intelligibility and distinctiveness (Dörnyei, 2014). For example, skills such as asking AI to generate outlines (Su et al., 2023), to generate the overall structure (Zou & Huang, 2024) and to organize the main points of writing logically (Kim et al., 2024) were combined into one strategy, i.e., “generating outline”. These helped us reduce the item number to 41, with 20 task processing strategies and 21 AI application regulatory strategies.

These 41 items were then subjected to three rounds of trials at a comprehensive university in China. In the first trial, 43 students were invited to: (1) read and rate each strategy based on their usage frequency; (2) indicate confusing or overlapped items and write down specific reasons; (3) write down strategies they used but not included in the list. Based on the responses, we further merged some items and rephrased some others for clearer meaning (e.g., “I evaluate the usefulness of AI tools’ output against my own standards” was changed into “I evaluate the usefulness of AI tools’ output against my knowledge or expectation”). Some new items were also added (e.g., “I use AI tools to generate PowerPoint slides”). After modification, the inventory was trialed with another 18 students, which led to some minor revisions in wording. The third trial with 11 students suggested no need for further modification.

The preliminarily finalized items, which consisted of 19 AI-assisted task processing strategies and 17 AI application regulatory strategies, were then incorporated into a questionnaire for subsequent validation. Given that SRL strategies are domain-specific and thus should be best measured at the course or even task level (Hsiao & Oxford, 2002; Pintrich, 2004), all the strategies in the questionnaire were set in the scene of learning for English language courses¹ at the university level. They were also rated on a five-point frequency scale (i.e., “never”, “rarely”, “sometimes”, “often” and “always”, which were coded as 1–5 in subsequent data analyses). Meanwhile, the questionnaire included questions about participants background information, such as gender, age, academic discipline, frequency of using AI and the number of AI tools used.

The validation involved two online questionnaire surveys. The first survey and its data analyses were intended for exploring the underlying structure of the inventory. The identified underlying structure was then further tested in the second survey. A simple trick question was inserted in the middle of the questionnaire to identify non-cooperators, whose responses were considered invalid if they answered this question wrong.

¹ The reasons for targeting these courses include: 1) English as a foreign language is a mandatory subject throughout the education system across China, ranging from primary through to tertiary levels, 2) most generative AI tools, at the time of data collection, were trained on large language models and thus were mostly used for processing or generating language-based information; 3) compared with many tasks in STEM fields like mathematics, language learning tasks at the tertiary level such as reading and essay writing tend to be more process-oriented (involving iterative processes like planning, drafting, revising and refining), value-laden and susceptible to subjective judgement, which render the use of AI-assisted task processing strategies at different stages and AI regulatory strategies such as evaluating AI-generated content particularly pertinent.

Participants

In the first survey, 553 undergraduate EFL (English as a Foreign Language) students studying at the aforementioned university in China participated in the questionnaire survey. After eliminating invalid responses, 488 students' data were retained. These 488 students aged between 16 and 23, with an average of 18.96. Among them, 232 (47.5%) were males and 256 (52.5%) were females; meanwhile, 200 (41%) were studying in the areas of arts, humanities and social science (ASH), while 288 (59%) were studying in areas of science, technology, engineering and mathematics (STEM). 73.0% of them also reported they "often" or "always" used AI tools for academic purposes, while only 27.0% reported they "sometimes" or "rarely" used such tools.

A total of 874 undergraduates from 13 universities located in eight cities of China participated in the second survey. Among them, only 707 students' responses were considered valid. These 707 respondents aged between 17 and 25, with an average age of 19.93. There were 381 (53.9%) males and 326 (46.1%) females, and 299 (42.3%) were studying AHS majors, while 408 (57.7%) were studying STEM majors. These students also had high frequency of using AI tools for learning, with 58.1% choosing "often" and 17.1% choosing "always".

Data collection

The two questionnaire surveys were conducted online in the October and December of 2024 respectively. In both surveys, a QR code of the questionnaire was shared with students by their teachers. Students interested in participation then accessed the questionnaire by scanning the code using their digital devices. Respondents were also instructed to carefully read the cover page of the questionnaire, which detailed the purpose and main content of the survey, the anonymity and voluntariness of participation, and information confidentiality. Last, they were instructed to click the "Agree" button if they agreed to participate and to exit the questionnaire if otherwise.

Data analyses

Before data analyses, data from each survey was checked for univariate and multivariate normality. Afterwards, exploratory factor analysis (EFA) was conducted in SPSS 24.0 with the first survey's data to identify potential latent structure. Specifically, principal axis analysis would be employed if the data did not meet multivariate normality assumption (i.e., Mardia's normalized multivariate kurtosis outside the ± 3 range) (Bentler, 2006; Fabrigar & Wegener, 2012). Given that many EFA methods tend to under-extract factors (Courtney, 2013), Rasch analysis was additionally performed in Winsteps (Linacre, 2021) to check the unidimensionality of each factor suggested by EFA, through examining decomposed residual variance. First contrast residual variance less than 2.0 would confirm the factor's unidimensionality, indicating no additional dimension can be extracted from the investigated factor. Item fit statistics such as infit and outfit mean-squares (MNSQs) were also inspected, with MNSQs above 1.5 flagging potential misfitting items and those below 0.5 signaling overfitting items (i.e., psychometrically redundant items) (Linacre, 2021). Meanwhile, each factor's internal consistency was examined by calculating Cronbach's α , with values above 0.70 suggesting acceptable reliability (Hair et al., 2014). All these analyses provided statistical evidence for evaluating subscale and item performance, but removal or retention of items was also determined by judgement of its

potential impact on subscale's cohesion and content validity, which was made through consensus among the three researchers.

The factor structure identified through the first stage was then further validated by fitting it to the data collected in the second survey, through means of confirmatory factor analysis (CFA) in the program R 4.4.2. Robust maximum likelihood estimator was chosen for parameter estimation if there was violation of multivariate normality assumptions (Bentler, 2006; Kline, 2016). In evaluating the overall fit of measurement model, we chose goodness-of-fit indices less sensitive to large sample sizes, including the χ^2/df ratio, standardized root mean square residual (SRMR), the root mean square error of approximation (RMSEA), the comparative fit index (CFI) and the Gamma hat; a non-significant χ^2/df ratio (i.e., less than 3.83), SRMR and RMSEA below or equal to 0.08, as well as CFI and Gamma hat equal to 0.90 or above indicate acceptable fit (Bentler, 1990; Fan & Sivo, 2007; Hu & Bentler, 1999; Marsh et al., 2004). Meanwhile, potential competing models with theoretical plausibility were tested and compared with the model identified through EFA, and Akaike's (1987) information criterion (AIC) was applied to judge model superiority: $\Delta AIC < 2$ indicates no significant difference between the compared models; a $\Delta AIC > 2$ suggests that the model with the smaller AIC is statistically superior (Burnham & Anderson, 2004).

Confirmed factors' average variance extracted (AVE) and heterotrait-monotrait ratio (HTMT) were examined for their convergent and discriminant validity, with $AVE \geq 0.50$ indicating good convergent validity (Brown, 2015; Hair et al., 2014) and $HTMT < 0.85$ suggesting good discriminant validity (Henseler et al., 2015). To investigate the equivalence of confirmed measurement model(s) across subgroups of the sample, we conducted multi-group CFA respectively across the two gender groups and the two broadly divided disciplinary groups, i.e., AHS versus STEM. Tests of configural (i.e., factor structure), metric (i.e., factor loadings), scalar (i.e., indicator intercepts) and residual (i.e., indicator residual variances) invariance were performed progressively, and invariance is confirmed if goodness-of-fit indices are acceptable and the difference in CFI between adjacent tests (ΔCFI) ≤ 0.01 (Brown, 2015; Cheung & Rensvold, 2002).

To examine the relationships between students' use of each type of strategies and general exposure to AI (including frequency of using AI and number of AI tools used), composite scores were calculated for each latent variable, and Pearson's bivariate correlation coefficients (r) were calculated, with an r between 0.10 and 0.30 indicating small correlation, that between 0.30 and 0.50 indicating medium correlation, and that above 0.50 indicating large correlation (Cohen, 1992). Last, to explore potential group differences in using those eight types of strategies, independent-samples t tests were conducted. Effect sizes (d) of differences were also computed, with a d between 0.20 and 0.40 indicating small difference, that between 0.40 and 0.60 indicating medium difference, and that above 0.60 representing large difference (Hattie, 2009).

Major findings

Findings from Survey I

Data screening indicated that the items were generally normally distributed. However, Mardia's normalized multivariate kurtosis was 42.50, suggesting significant deviation from multivariate normal distribution (Bentler, 2006). Due to this, subsequent EFA was conducted through principal axis analysis, a method that does not assume assumption

of multivariate normality (Fabrigar & Wegener, 2012). The Kaiser–Meyer–Olkin Measures was 0.918, with a significant Bartlett’s Test of Sphericity ($p < 0.001$). Principal axis analysis with Oblimin rotation suggested an 8-factor solution. This 8-factor solution accounted for a cumulative 64.85% of variance. After closer examination of item content, the 8 factors could be meaningfully interpreted as follows:

- 1) *Planning & performing* (PP), which concerns using AI to plan, to search for and process source material, and to generate ideas during task processing (e.g., “I ask AI tools to break down a complex task into manageable steps”, “I use AI tools to summarize the main points in reading material”);
- 2) *Product refinement* (PR), which refers to using AI to refine the language and content of one’s own product (e.g., “I use AI tools to polish the language of my assignment (e.g., word choice, tone, style, etc.)”);
- 3) *Evaluation & assessment* (EA), which refers to using AI to evaluate or assess one’s own product according to task requirements or standards (e.g., “I use AI tools to evaluate my assignment against the requirements or scoring criteria of the task”);
- 4) *Language compensation* (LC), which is about using AI to bridge linguistic gaps between languages via translation or meaning querying (e.g., “I use AI tools to translate English material into Chinese for better understanding”);
- 5) *Output-focused metacognitive regulation* (OMR), which comprises AI output focused metacognitive strategies, such as designing prompts (i.e., prompt engineering), monitoring or evaluating AI output and adjusting prompts (e.g., “I evaluate the usefulness of AI tools’ output against my knowledge or expectation”);
- 6) *Tool-focused metacognitive regulation* (TMR), which concerns AI tool focused metacognitive strategies, such as purposefully selecting AI tools, appraising their capacity, and adapting to tool failures by looking for new tools (e.g., “I purposely select AI tools for different learning tasks”);
- 7) *Behavioral regulation* (BR), which refers to managing and dedicating one’s effort to learning new knowledge and skills about AI application through different means, such as self-exploration, learning from peers or online resources, etc. (e.g., “I exchange ideas with my friends about how to use AI tools to help learning”);
- 8) *Ethical regulation* (ER), which refers to self-imposing restrictions on AI application out of ethical considerations for issues such as plagiarism and negative ramifications (e.g., “I remind myself of the potential negative influence of inappropriately using AI tools on academic integrity (e.g., originality, honesty, fairness, etc.)”).

It was also found that Item 14 (“I use AI tools to generate PowerPoint slides”), which seems to belong to PP in content, loaded on EA. Meanwhile, Rasch analyses disclosed that this item had infit and outfit MNSQ of 1.64 and 2.00 respectively, suggesting significant misfit (Linacre, 2021). Considering factor cohesion and subscale reliability, this item was dropped and the data were re-analyzed. The results are presented in Table 1.

Table 1 also displays the Rasch analyses results. We can see that all factors had first contrast residual variance below 2.0, confirming their unidimensionality; also, all items’ infit and outfit MNSQs fell within the 0.5–1.5 range, suggesting overall good fit of those items (Linacre, 2021).

Table 1 Results of exploratory factor and Rasch analyses (n = 488)

Factor name (Cronbach's α)	Item	Exploratory factor analysis		Rasch analysis			
		Factor loading	Eigenvalue (% of variance)	Measure	Infit MNSQ	Outfit MNSQ	1st contrast
Planning and performing ($\alpha = 0.87$)	06	0.643	1.44 (4.10%)	−0.19	0.78	0.79	1.80
	03	0.603		−0.20	0.75	0.75	
	07	0.594		0.37	0.89	0.90	
	08	0.564		0.01	1.00	1.05	
	02	0.518		−0.16	1.01	1.17	
	11	0.501		0.73	0.98	1.01	
	10	0.490		−0.29	1.02	1.02	
	09	0.380		−0.53	1.20	1.16	
	01	0.350		0.26	1.34	1.43	
	16	0.784		0.04	0.73	0.73	
Product refinement ($\alpha = 0.83$)	17	0.548	1.14 (3.25%)	0.45	1.05	1.06	1.70
	15	0.543		−0.50	1.22	1.15	
Evaluation and assessment ($\alpha = 0.86$)	18	0.812	1.23 (3.50%)	0.06	1.06	1.02	1.60
	12	0.736		−0.75	0.83	0.88	
	19	0.723		0.69	1.08	1.00	
Language compensation ($\alpha = 0.73$)	04	0.681	1.67(4.78%)	−0.24	0.88	0.84	1.70
	13	0.675		0.45	0.93	0.93	
	05	0.560		−0.22	1.17	1.17	
Output-focused meta-cognitive regulation ($\alpha = 0.80$)	22	0.748	3.19 (9.22%)	−0.22	1.02	0.97	1.40
	27	0.595		−0.07	0.91	0.86	
	21	0.549		0.68	1.03	1.07	
	28	0.508		−0.67	0.94	0.92	
	23	0.459		0.29	1.16	1.22	
Tool-focused metacognitive regulation ($\alpha = 0.82$)	25	0.818	1.02 (2.90%)	−0.01	0.74	0.72	1.50
	24	0.607		0.58	1.06	1.10	
	29	0.511		−0.42	1.18	1.10	
	26	0.417		−0.26	0.84	0.82	
	20	0.320		0.11	1.20	1.26	
Behavioral regulation ($\alpha = 0.83$)	32	0.695	11.04 (31.53%)	−0.29	0.90	0.88	1.70
	33	0.667		−0.25	0.88	0.88	
	30	0.570		−0.14	1.00	0.98	
	31	0.491		0.69	1.21	1.23	
Ethical regulation ($\alpha = 0.83$)	36	0.888	1.98 (5.67%)	0.09	0.79	0.82	1.60
	34	0.756		−0.06	1.14	1.11	
	35	0.734		−0.03	1.04	1.03	

Findings from Survey II

After removing Item 14, the questionnaire was administered to undergraduates from 13 universities (valid $n = 707$). Data from this second survey were also generally normally distributed at the item level (see Appendix), but violated the assumption of multivariate normal distribution, as Mardia's normalized multivariate kurtosis was 84.11 (Bentler, 2006; Kline, 2016). Therefore, robust maximum likelihood estimator was chosen to cope with such violation when conducting CFA to test the 8-factor model identified in EFA. Additionally, we tested two competing models (see Fig. 2): a 2-factor model positing that items from the two levels of regulation (i.e., regulation of AI-assisted task processing and AI application) were explained by two distinct though inter-correlated factors; a 1-factor model assuming that all items could be directly accounted for by a general factor.

Fit statistics of the three models are summarized in Table 2. Only the 8-factor model achieved acceptable fit, as its χ^2/df ratio was below 3.83, SRMR and RMSEA (including

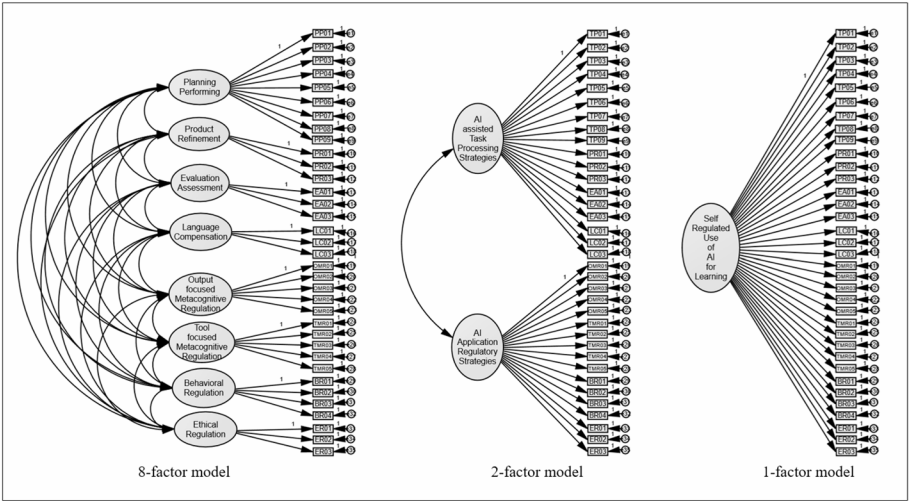


Fig. 2 Hypothesized measurement models

Table 2 Goodness-of-fit of measurement models (n = 707)

Model	χ^2	df	χ^2/df	SRMR	RMSEA (90%CI)	CFI	Gamma hat	AIC
8-factor model	1310.82	532	2.46	0.05	0.05 (0.04–0.05)	0.92	0.94	64,046.44
2-factor model	3320.55	559	5.94	0.08	0.08 (0.08–0.09)	0.71	0.82	66,591.06
1-factor model	4517.64	560	8.07	0.10	0.10 (0.10–0.10)	0.59	0.76	68,163.95

SRMR, Standardized root mean residual; RMSEA, Root mean square error of approximation; CFI, Comparative fit index; AIC, Akaike information criterion

its confidence intervals) were both below 0.08, and CFI and Gamma hat were both above 0.90, whereas the other two models did not meet these criteria (Bentler, 1990; Fan & Sivo, 2007; Hu & Bentler, 1999; Marsh et al., 2004). Moreover, this model’s AIC was significantly smaller than those of the other two, confirming its superiority.

Figure 3 is a graphic illustration of the 8-factor model with standardized factor loadings and correlations. The factor loadings ranged from 0.60 to 0.90 and were all statistically significant at the 0.001 level. All factors had AVEs above 0.50, except for *behavioral regulation*, which had an AVE of 0.49 (see Table 3). These indicate that those factors generally had good convergent validity (Brown, 2015; Hair et al., 2014). The HTMT ratios were all below 0.85, suggesting good discriminant validity among factors (Henseler et al., 2015). Last, Cronbach’s α of each factor ranged from 0.77 to 0.90, indicating overall good reliability (Hair et al., 2014).

The 8-factor model also showed measurement invariance between both the two gender groups and the two academic disciplinary groups (see Table 4), as SRMR and RMSEA were all below 0.08 and CFI was consistently above 0.90 when tests of configural, metric, scalar and residual invariance were progressively performed; the changes in CFI value (Δ CFI) were also within the ± 0.01 range (Cheung & Rensvold, 2002).

To account for the generally moderate to large inter-correlations within each level of regulation, we further tested a second-order-factor model assuming that factors from the same level of regulation were explained by a common second-order factor, which was also correlated with the other second-order factor. This model also had acceptable fit to the data ($\chi^2/df=2.59$, SRMR=0.06, RMSEA=0.05, 90% CI=0.04–0.05, CFI=0.91, gamma hat=0.94). Figure 4 shows its standardized coefficients. The second-order factors

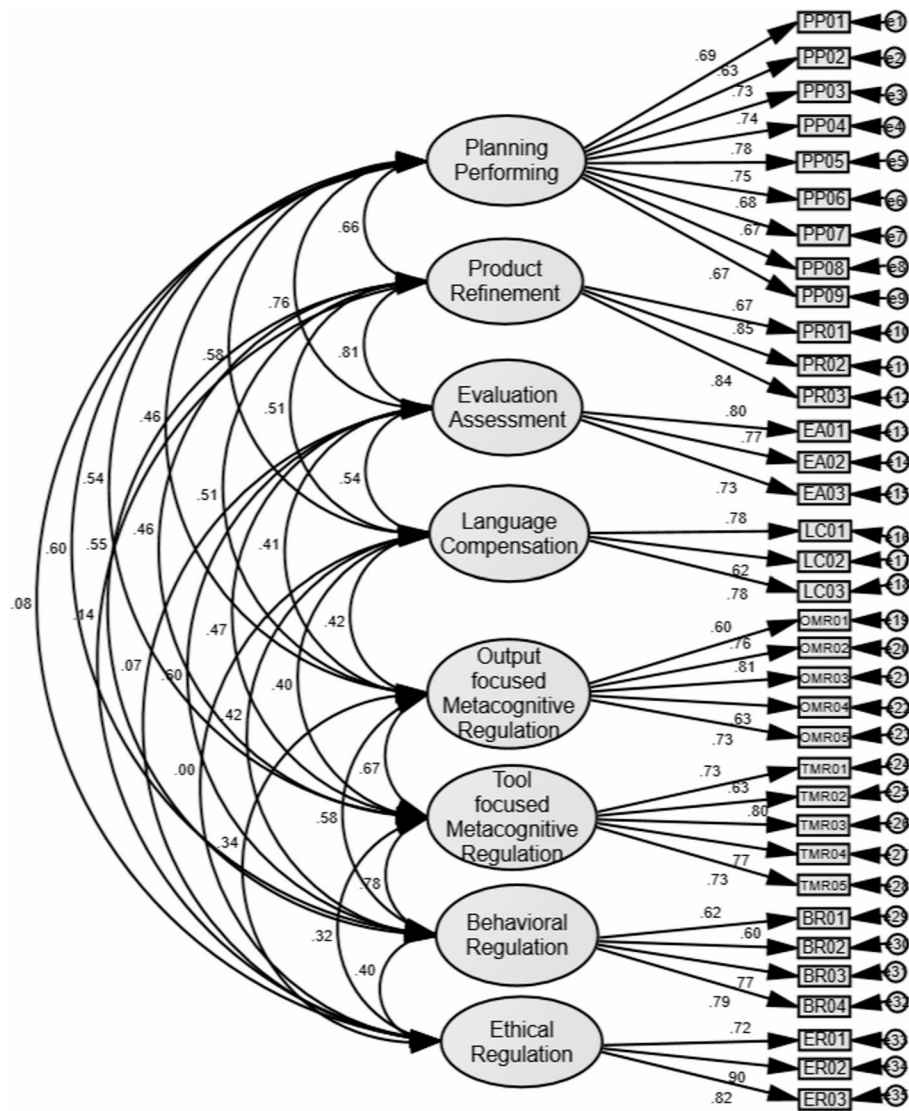


Fig. 3 8-Factor measurement model with standardized coefficients

Table 3 Factors' reliability, convergent and discriminant validity (n = 707)

Factor	α	AVE	Heterotrait-monotrait ratio (HTMT)						
			PP	PR	EA	LC	OR	TR	BR
PP	0.90	0.50							
PR	0.83	0.62	0.654						
EA	0.81	0.59	0.762	0.801					
LC	0.77	0.53	0.587	0.538	0.547				
OMR	0.84	0.53	0.460	0.525	0.411	0.415			
TMR	0.85	0.54	0.550	0.481	0.475	0.428	0.689		
BR	0.79	0.49	0.614	0.567	0.607	0.460	0.591	0.774	
ER	0.84	0.66	0.092	0.159	0.077	0.076	0.375	0.318	0.428

PP, Planning & performing; PR, Product refinement; EA, Evaluation & assessment; LC, Language compensation; OMR, Output-focused metacognitive regulation; TMR, Tool-focused metacognitive regulation; BR, Behavioral regulation; ER, Ethical regulation; AVE, Average variance extracted

Table 4 Tests of measurement invariance across gender and discipline

Grouping	Model	χ^2	df	χ^2/df	SRMR	RMSEA	CFI	ΔCFI
Gender: male (n = 381); female (n = 326)	Configural	1881	1064	1.77	0.052	0.047	0.916	-
	Metric	1912	1091	1.75	0.053	0.046	0.915	-0.001
	Scaler	1981	1118	1.77	0.054	0.047	0.911	-0.004
	Residual	2034	1153	1.76	0.054	0.047	0.909	-0.002
Discipline: AHS (n = 299); STEM (n = 408)	Configural	1910	1064	1.80	0.052	0.047	0.912	-
	Metric	1938	1091	1.78	0.052	0.047	0.912	0.000
	Scaler	2015	1118	1.80	0.053	0.048	0.907	-0.005
	Residual	2046	1153	1.77	0.054	0.047	0.907	0.000

SRMR, Standardized root mean residual; RMSEA, Root mean square error of approximation; CFI, Comparative fit index; AHS, Arts, humanities & social science; STEM, Science, technology, engineering & mathematics

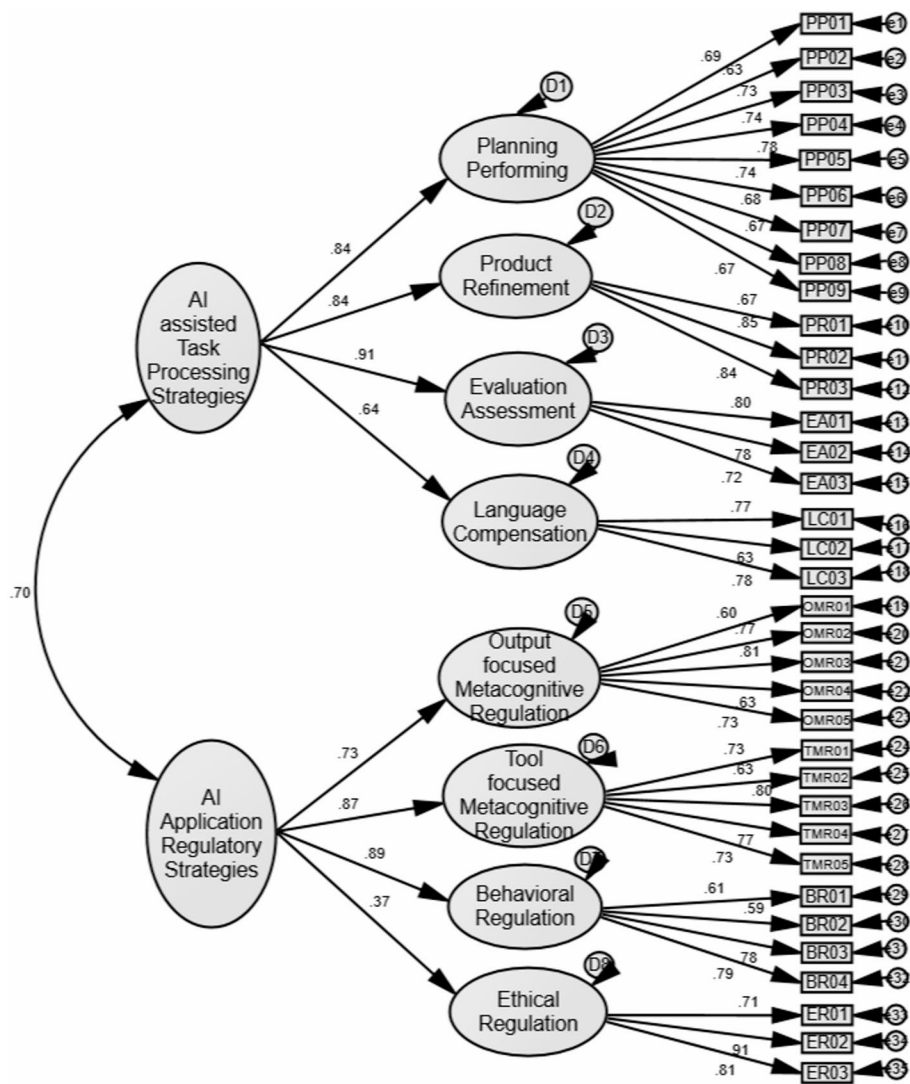


Fig. 4 Second-order-factor model with standardized coefficients

had a correlation of 0.70, suggesting the two levels of regulation are closely related but psychometrically distinguishable from each other.

Strategy usage frequency and the roles of gender, academic discipline and AI exposure

To gauge students' average frequency of using the eight types of AI strategies, each factor's composite score was calculated by averaging item scores within the same factor. As illustrated in Fig. 5, among AI-assisted task processing strategies, *evaluation & assessment* was the least frequently used strategy type ($M = 2.99$), while the other three types had very similar average frequencies ($M = 3.31 - 3.40$). Among AI application regulatory strategies, *output-focused metacognitive regulation* was most frequently employed ($M = 3.82$), whereas *behavioral regulation* had the lowest frequency ($M = 3.34$).

Independent samples t tests revealed that, compared with male participants, female participants reported statistically significantly more frequent uses of both *tool-focused metacognitive regulation* (female: $M = 3.64$, $SD = 0.85$; male: $M = 3.48$, $SD = 0.92$; $t = -2.46$, $df = 705$, $p = 0.014$) and *ethical regulation* (female: $M = 3.82$, $SD = 0.87$; male: $M = 3.49$, $SD = 1.01$; $t = -4.63$, $df = 705$, $p = 0.000$), though effect sizes were small ($d = 0.18$ and 0.35 respectively). Between the two academic disciplinary groups, the AHS group ($M = 3.76$, $SD = 0.93$) reported more frequent usage of *ethical regulation* than the STEM group ($M = 3.55$, $SD = 0.98$), and the difference was statistically significant with a small effect size ($t = 2.98$, $df = 705$, $p = 0.003$, $d = 0.22$). No other strategies differed significantly across gender or discipline.

Table 5 displays the correlations between AI exposure and strategy usage. Students' general AI usage frequency had consistently moderate correlations with most strategy types ($r = 0.328 - 0.391$), except for a large correlation with *planning & performing* ($r = 0.549$) and no correlation with *ethical regulation* ($r = 0.000$). The number of AI tools used exhibited small-to-medium correlations across most strategies ($r = 0.098 - 0.334$), with the strongest association observed for *tool-focused metacognitive regulation* ($r = 0.334$). *Ethical regulation* showed no statistically significant relationship with this AI exposure metric again ($r = 0.033$).

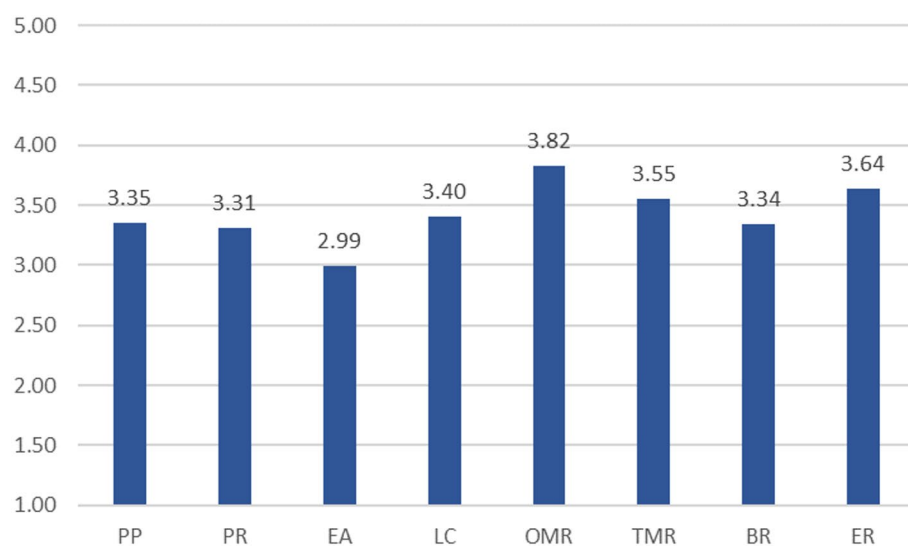


Fig. 5 Average usage frequency by strategy type

Table 5 Correlations between AI usage features and AI strategies (n = 707)

	Frequency of using AI	Number of AI tools used
Planning and performing	0.549**	0.190**
Product refinement	0.391**	0.237**
Evaluation and assessment	0.386**	0.162**
Language compensation	0.328**	0.098**
Output-focused metacognitive regulation	0.382**	0.251**
Tool-focused metacognitive regulation	0.337**	0.334**
Behavioral regulation	0.358**	0.225**
Ethical regulation	0.000	0.033
Number of AI tools used	0.205**	1.000

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Discussion

Factor structure of self-regulatory strategies of using AI for learning

Factorial analyses of the two surveys' data confirmed an eight-factor structure for the developed instrument. These factors demonstrated overall good psychometric properties, such as unidimensionality, convergent and discriminant validity, reliability and measurement invariance across gender and discipline. Hierarchical modeling further categorized these factors into two higher-order constructs: *AI-assisted task processing strategies* and *AI application regulatory strategies*. This empirically validates our hypothesis that strategies for regulation of AI application (e.g., planning, monitoring and adjusting AI prompts and AI tool usage) are distinct from those for regulation of AI-assisted task processing (e.g., generating AI feedback on task performance), extending frameworks of traditional SRL (Boekaerts, 1999; Pintrich, 2004; Zimmerman, 2002) into the AI-empowered learning context. It also highlights the added complexity of AI-empowered SRL, which is not a simple addition of AI tools into traditional SRL processes but involves more complex processes of tool-oriented planning, appraisal, upskilling and ethical regulation (Abdelhalim, 2024; Bearman et al., 2024; Jiang, 2024; Liu et al., 2024; Ou et al., 2024; Tam, 2024; Xiao et al., 2025; Zou & Huang, 2024).

Among AI-assisted task processing strategies, *planning & performing* and *product refinement* align with the forethought and performance (or monitoring and control) stages of SRL (Pintrich, 2004; Zimmerman, 2002), during which students plan, deploy, monitor and adjust various cognitive strategies to address learning tasks. The last dimension, i.e., *evaluation & assessment*, corresponds to the reflection stage of SRL, where students evaluate their own task performance (Pintrich, 2004; Zimmerman, 2002). Notably, although similar to conventional cognitive and metacognitive SRL strategies taken by face value, these strategies actually diverge from them by relying on AI mediation. For example, within traditional framework of SRL, learners rely on their own initiative to monitor their task progress and evaluate task performance in a self-reflective manner, while AI technology now allows for automatic completion of such processes. To what extent such technologically scaffolded or even automated metacognitive strategies still tap into students' metacognitive resources warrants further investigation.

Meanwhile, the literature on AI-enhanced learning strategies witnesses a significant amount of findings on using AI to address linguistic gaps, particularly among non-native speakers when dealing with tasks in the target language (Abdelhalim, 2024; Guo & Li, 2024; Kim et al., 2024; Ou et al., 2024; Yan, 2023; Zou & Huang, 2024). In keeping with these findings, the current study identified *language compensation* as a distinct factor

beyond those stage-aligned strategies. In fact, this group of strategies were reported as the most frequently utilized among the four types of task processing strategies by the Chinese respondents, indicative of their pronounced need to overcome linguistic barriers when completing tasks in English.

Although the extent to which AI-empowered “metacognitive” task-processing strategies embody genuine metacognition remains unclear, the metacognitive regulation of AI application identified in the literature and in the present study does capitalize on students’ proactive planning, monitoring, controlling and evaluation of AI utilization, and thus are essentially metacognitive. Moreover, our study found that metacognitive regulation of AI *output* and AI *tools* were perceived by the participants as distinct constructs, shedding fresh light on this aspect of AI application. Indeed, managing the process of using AI output can be rather different from regulating the process of using AI tools, as the former focuses on the goal setting, elicitation, monitoring and actions upon specific AI-generated content, while the latter requires users to plan, apply and appraise the overall performance of AI tools in addressing learning needs. In Xiao et al.’s (2025) qualitative study, students also frequently reported their assessment of specific AI output during their interactions with those tools, while evaluating the performance of AI tools required them to holistically reflect on their cumulative experiences with those tools, rather than focusing on isolated output.

Furthermore, *behavioral* and *ethical regulation* were identified as another two distinguishable constructs. According to Flavell (1979), metacognitive knowledge includes one’s knowledge about self, task and strategies. As a novel strategy for learning, efficient application of AI technology becomes new knowledge to be acquired by novice users, and thus whether being able to learn from diverse sources about AI utilization and update one’s AI literacy is integral to self-regulated use of AI. Such agentive effort control parallels Pintrich’s (2004) behavior regulation, wherein learners actively manage their time, persistence and social behaviors to attain learning goals (Teng & Zhang, 2016; Wang & Bai, 2017). Beyond operational skills, self-regulated AI application also entails an awareness of using it responsibly and ethically, given the potential societal ramifications of misuse (Long & Magerko, 2020; Ng et al., 2023) and presence of usage policies in many contexts (Moorhouse et al., 2023). There is also ample qualitative evidence showing students develop such awareness while using AI for learning, which subsequently shapes their AI-related behaviors (Ou et al., 2024; Tossell et al., 2024; Yan, 2023; Zou & Huang, 2024). Therefore, ethical regulation represents a crucial aspect of users’ interaction with the social context of AI applications.

Frequency of strategy use and the roles of gender, academic discipline and AI exposure

Among the eight strategy types, *output-focused metacognitive regulation* had the highest mean, suggesting its prevalence among the participants. This is likely due to the fact that AI-assisted task processing frequently necessitates evaluation of AI-generated output (Bearman et al., 2024; Liu et al., 2024; Ou et al., 2024; Tam, 2024), which serves as the basis for more advanced metacognitive processes, such as appraising the capacity of specific AI tools and AI technology broadly (Xiao et al., 2025). However, Bearman et al. (2024) caution that frequent evaluations of AI output does not inherently cultivate evaluative competence, which requires both domain expertise and familiarity with

diverse assessment or evaluative procedures, and conscious training of such procedures are crucial.

In contrast to the high reported frequency of self-initiated AI output evaluation, participants employed AI tools to assess their own ideas and assignments far less frequently, as evidenced by the lowest mean for *evaluation & assessment*. The underutilization of such formative strategies may relate to Chinese students' overall prioritization of summative assessments, such as high-stakes university entrance exams, over formative assessments like peer- and self-assessment, leading to less focus on meaningful self-reflection and feedback-seeking in favor of achieving high test scores (Li et al., 2021; Yan, 2016). This carries important pedagogical implications, as prior studies demonstrated that, compared with using AI to improve assignment quality in a fast, convenient and utilitarian manner, AI tools can be more meaningfully leveraged to evaluate, assess and provide useful feedback on task performance (Su et al., 2023; Tam, 2024). Truly self-regulated learners would actively engage with such feedback to enhance their competence (Abdelhalim, 2024; Su et al., 2023). Therefore, explicit instruction on how to engage with AI for formative assessment of learning and task performance is needed.

Our study also revealed that female students engaged more frequently in *tool-focused metacognitive regulation* and *ethical regulation* than male students. The difference in using the former type of strategies may stem from differential AI tool exposure, as females ($M=2.68$, $SD=1.35$) reported using more numbers of AI tools on average than males ($M=2.30$, $SD=1.50$; $t=-3.53$, $df=705$, $p=0.000$), indicating their exposure to a greater variety of those tools. The gender gap in *ethical regulation* mirrors findings by Wang et al. (2025) and Jang et al. (2022), which also discovered that female students tended to be more ethically aware regarding using AI for general purposes. Such gender difference can be accounted for by earlier theories of moral development (Gilligan, 1993) and social role (Eagly, 2013), which posit that during their socialization process, women tend to adopt community-centered roles and care-oriented ethic, emphasizing relationships, empathy and responsibility to others, while men lean toward agentic roles and justice-oriented ethic, focusing on rules, autonomy and individual rights. When it comes to AI use, therefore, females might be more sensitive than males to its societal ramifications, including its potential to jeopardize trust, fairness and academic integrity.

In terms of academic discipline, our study found that AHS students engaged more frequently than STEM students in *ethical regulation*. This may be explained by the different educational focus between the two broad fields. According to Snow's (2012) theory of the "two cultures", humanistic fields emphasize interpretive, reflective and value-oriented thinking, while science fields prioritize empirical, technical, and solution-driven methodologies. This divide may have caused the differential epistemic approaches to technology such as AI, with the former group focusing more on human experience, emotions and morality issues incurred by technology use, while the latter more on the utility and efficiency of technology itself. However, Wang et al. (2025) did not find such difference between the two groups. This discrepancy in finding may reflect contextual variation, as Wang et al.'s (2025) sample included participants from Hong Kong, where ethical education's disciplinary coverage may differ from that in Mainland China. Another cause might be associated with difference in measurement focus, as Wang et al.'s (2025) scales assess general AI ethics, whereas our ethical regulation subscale focuses on ethical reflection in academic settings. These aside, findings from these studies highlight

the need for targeted interventions to address gender and disciplinary gaps in strategy education.

Difference in measurement focus may have caused another gap in findings: while Wang et al. (2025) found a positive relation between students' AI exposure and ethics reflection, our study found no association between them. Our study's finding suggests that mere exposure does not spontaneously foster ethical regulation in the academic context, and thus explicit instruction is necessary. Meanwhile, our study disclosed that AI exposure statistically significantly correlated with other strategies. Particularly, *planning & performing* exhibited a strong relationship with general AI usage frequency, emphasizing its centrality in student-AI interaction. *Tool-focused metacognitive regulation* correlated most strongly with the number of AI tools used. This is likely because engaging with more AI tools facilitates comparative evaluation and selection among tools, which are strategies essential to this construct. This also implies that to foster the usage of this type of strategies, teaching activities can be designed to encourage application of multiple tools.

Fostering metacognitive and ethical awareness of regulating AI application

The above findings suggest that it is necessary to treat AI-assisted task processing and regulation of AI application as distinguishable yet interconnected dimensions of AI-empowered SRL. This underscores the need for balanced development of both competencies among students. Of particular importance is fostering students' metacognitive and ethical awareness of regulating their AI application process.

Various studies have revealed that students often blindly accept AI-generated content and bear misconceptions about AI's capabilities in tackling learning tasks (Abdelhalim, 2024; Amaro et al., 2023; Tam, 2024; Xiao et al., 2025; Yan, 2023; Yang et al., 2024). To address this, it is necessary for them to develop a systematic knowledge base about AI's features, such as strengths and weaknesses, as well as *when, where, why* and *how* to use particular AI tools to help meet specific learning needs (Xiao et al., 2025). Cultivating such metacognitive knowledge involves critically evaluating AI-generated output, appraising the functionality or utility of the AI tools chosen, and reflecting on the effectiveness of the ways they were used (such as prompting strategy and interaction process); meanwhile, students can benefit from proactively learning about AI's new features and use cases through peer discussions, online resources and hands-on experimentation with new tools and features (Xiao et al., 2025; Yan, 2023).

On the other hand, many students are found to use AI tools in unethical or harmful ways, such as relying on them for ghostwriting or generating superficial answers (Nguyen & Goto, 2024; Yeo, 2023). Given that students' ethical awareness does not automatically develop with their increased engagement with AI, as findings from the present study indicate, concerted efforts from teachers, institutions, and the broader education system are essential. Specifically, measures such as clear guidelines, tutorial, workshops and lectures should be put in place to instill responsible AI practices among students. Explicit instructional interventions can also be integrated into curriculum design, such as embedding dedicated modules on AI ethics within core subjects that incorporate real-world case studies of ethical dilemmas, role-playing exercises to simulate decision-making scenarios, and mandatory assignments requiring students to document and justify their AI usage ethically.

Potential uses of the instrument

The validated AI strategy inventory has potential applications across multiple scenarios among educational stakeholders, including serving as diagnostic, instructional, and evaluative instruments, fostering data-driven decision-making and enhancing the integration of AI into education. For learners, this inventory provides a structured framework for self-assessment, enabling reflection on their AI application approach. By evaluating strategy use, students can identify gaps in their approach to leveraging AI for addressing academic tasks, and adjust their practices accordingly. For example, the first part of the inventory (i.e., AI-assisted task processing strategies) helps learners recognize whether their AI usage supports a process-oriented and reflective approach (e.g., sufficient use of *planning & performing* and *evaluation & assessment* strategies) or risks over-reliance on shortcuts to task completion (e.g., excessive use of *product refinement* strategies). Such insights empower students to balance technological dependency with intentional skill-building. The second part (i.e., AI application regulatory strategies), moreover, encourages learners to critically assess if and to what extent they plan, monitor, and adapt their AI application process. This helps them develop self-awareness of proactively controlling their AI use and become strategically efficient and ethically responsible AI users.

Teachers can also employ this inventory to diagnose students' AI strategy use in formative assessment, informing differentiated instruction. By analyzing patterns in strategy deployment, teachers may tailor feedback, recommend targeted AI tools, or design activities that scaffold ethical and effective AI integration. For instance, if metacognitive strategies are under-utilized, teachers might incorporate reflective exercises on AI tool selection and output evaluation. Furthermore, professional development programs could utilize this inventory to train educators in fostering AI literacy, ensuring they guide students in navigating AI's affordances and limitations.

For educational research, this instrument can be leveraged to investigate the effectiveness of AI-scaffolded learning. Quantitative analyses of strategy use may reveal correlations between specific AI practices and learning outcomes, informing theory-building around AI's role in learning. Comparative studies might explore demographic or contextual variations in strategy adoption (see e.g., Abdelhalim, 2024), such as differences between novice and advanced learners. Longitudinal designs could track how use of different AI strategies evolves with increasing AI engagement, offering insights into learners' adaptability. Such research not only advances knowledge about AI-enhanced learning but also provides actionable recommendations for practitioners and policymakers.

Conclusions

This study developed and validated an inventory of AI strategies in the academic context and identified eight underlying factors, with four pertaining to the regulation of task processing and another four concerning the regulation of AI application *per se*. The corresponding subscales demonstrated robust psychometric properties, supporting their reliability and validity. Analyses of the collected data revealed an imbalanced utilization of different AI-mediated strategies, with gender, academic discipline, and degree of AI exposure significantly associated with the frequency of certain strategy use. These findings, along with the proposed model, represent one of the first endeavors to delineate the components of strategic use of AI for learning and quantitatively investigate

students' usage of specific strategies. By synthesizing extant evidence on strategy use and analyzing empirical data, this study makes significant contributions to extending our knowledge of SRL in the context of AI, providing a foundation for future research on AI-integrated learning process. Moreover, the validated inventory can serve as a valuable tool for key educational stakeholders. Particularly, teachers and learners can employ this instrument to diagnose AI strategy usage in formative assessment, enabling identification of gaps and informing targeted interventions.

It is important to acknowledge that the inventory was validated within the specific context of learning English as a foreign language in China. Given potential task and skill differences across subjects as well as cultural differences in learning styles, further validation studies based on data collected within other academic disciplinary and cultural settings are needed to establish the generalizability of the identified factor structure. Additionally, future research could investigate the instrument's criterion and predictive validity by examining its correlations with other indicators of AI strategy use and AI-mediated learning outcomes. Despite these limitations, this pioneering study represents a significant advancement in the field of AI-mediated SRL and offers a valuable tool for AI-enhanced education.

Appendix

Code	Item	Survey I (n = 488)				Survey II (n = 707)			
		Mean	SD	Skew-ness	Kur-tosis	Mean	SD	Skew-ness	Kur-tosis
PP01	1. I ask AI tools to break down a complex task into manageable steps	3.02	1.29	−0.11	−1.13	2.78	1.15	0.27	−0.79
PP02	2. I use AI tools to search for material or resources	3.32	1.12	−0.44	−0.58	3.62	1.00	−0.60	−0.09
PP03	3. I use AI tools to find specific information in reading material	3.35	1.13	−0.48	−0.53	3.53	1.03	−0.55	−0.20
PP04	6. I use AI tools to summarize the main points in reading material	3.34	1.18	−0.39	−0.74	3.45	1.11	−0.52	−0.42
PP05	7. I use AI tools to integrate the content of different material	2.93	1.27	−0.01	−1.11	3.20	1.18	−0.28	−0.76
PP06	8. I use AI tools to generate summaries of online information about the assignment's topic	3.20	1.21	−0.36	−0.86	3.44	1.13	−0.52	−0.51
PP07	9. I ask AI tools about the definition of concepts involved in the assignment	3.58	1.13	−0.64	−0.31	3.59	1.05	−0.71	0.06

Code	Item	Survey I (n = 488)				Survey II (n = 707)			
		Mean	SD	Skew-ness	Kur-tosis	Mean	SD	Skew-ness	Kur-tosis
PP08	10. I use AI tools to brainstorm ideas (e.g., topics, subtopics, arguments, supporting evidence, etc.) for the assignment	3.42	1.18	−0.46	−0.69	3.41	1.13	−0.44	−0.58
PP09	11. I use AI tools to generate an outline for the assignment	2.66	1.25	0.34	−0.92	3.13	1.22	−0.23	−0.94
PR01	15. I use AI tools to check and correct the mechanical errors (e.g., grammar, spelling and punctuation mistakes) in my assignment	3.36	1.29	−0.42	−0.96	3.31	1.24	−0.31	−0.95
PR02	16. I use AI tools to polish the language of my assignment (e.g., word choice, tone, style, etc.)	3.13	1.30	−0.25	−1.06	3.38	1.16	−0.40	−0.70
PR03	17. I use AI tools to improve the content and organization of my assignment (e.g., logic, structure, coherence, etc.)	2.96	1.25	−0.14	−1.06	3.23	1.18	−0.26	−0.83
EA01	12. I ask AI tools for feedback on my own ideas when writing the assignment	2.97	1.22	−0.06	−0.96	3.15	1.21	−0.25	−0.91
EA02	18. I use AI tools to better understand the requirements or scoring criteria of the task	2.67	1.33	0.21	−1.17	3.12	1.22	−0.17	−0.99
EA03	19. I use AI tools to evaluate my assignment against the requirements or scoring criteria of the task	2.44	1.33	0.49	−1.01	2.70	1.25	0.19	−1.06
LC01	4. I use AI tools to translate English material into Chinese for better understanding	3.17	1.32	−0.29	−1.08	3.64	1.13	−0.69	−0.23
LC02	5. I use AI tools to search for the meaning of new words or expressions	3.16	1.27	−0.19	−1.08	3.32	1.24	−0.33	−0.93
LC03	13. I use AI tools to translate my Chinese words into English when writing the assignment	2.76	1.33	0.15	−1.18	3.25	1.21	−0.33	−0.85

Code	Item	Survey I (n = 488)				Survey II (n = 707)			
		Mean	SD	Skew-ness	Kur-tosis	Mean	SD	Skew-ness	Kur-tosis
OMR01	27. I think about how to write prompts to get useful information from AI tools	3.98	1.03	−1.08	0.90	3.60	1.14	−0.64	−0.36
OMR02	21. I have specific expectations for the output from AI tools in handling learning tasks	3.61	1.07	−0.64	−0.08	3.72	0.98	−0.63	0.07
OMR03	22. I evaluate the usefulness of AI tools' output against my knowledge or expectation	4.05	1.06	−1.27	1.22	3.96	0.97	−0.97	0.77
OMR04	23. I fact-check AI tools' output against other sources	3.81	1.10	−0.82	0.00	3.68	1.07	−0.59	−0.35
OMR05	28. I would revise the prompts when I find AI tools' output is not satisfactory	4.23	0.81	−0.91	0.37	4.17	0.89	−1.17	1.43
TMR01	20. I purposely select AI tools for different learning tasks	3.36	1.21	−0.54	−0.68	3.57	1.17	−0.53	−0.60
TMR02	24. For the same topic or question, I ask different AI tools to generate answers and then compare those answers	3.07	1.26	−0.21	−0.99	3.32	1.20	−0.33	−0.81
TMR03	25. I compare different AI tools in terms of their advantages and disadvantages	3.44	1.22	−0.58	−0.64	3.48	1.14	−0.50	−0.56
TMR04	26. I evaluate the abilities of AI tools in handling different types of tasks	3.58	1.10	−0.74	−0.09	3.51	1.10	−0.53	−0.33
TMR05	29. When one AI tool fails me in handling a task, I would search for and use other AI tools	3.67	1.20	−0.79	−0.29	3.89	1.03	−1.01	0.71
BR01	30. I exchange ideas with my friends about how to use AI tools to help learning	3.37	1.20	−0.44	−0.72	3.35	1.03	−0.41	−0.36
BR02	31. I search for online resources (e.g., videos, posts, articles) about how AI tools can be used to support my learning	2.94	1.25	−0.04	−1.07	3.08	1.16	−0.21	−0.81

Code	Item	Survey I (n = 488)				Survey II (n = 707)			
		Mean	SD	Skew-ness	Kur-tosis	Mean	SD	Skew-ness	Kur-tosis
BR03	32. I try to find out new functions of the AI tools I have been using	3.44	1.19	−0.39	−0.75	3.45	1.04	−0.25	−0.58
BR04	33. I keep exploring new AI tools	3.42	1.21	−0.38	−0.82	3.49	1.08	−0.32	−0.66
ER01	34. I remind myself of what is not appropriate or allowed when using AI tools	3.78	1.12	−0.74	−0.16	3.41	1.22	−0.39	−0.80
ER02	35. I remind myself of the potential negative influence of inappropriately using AI tools on my learning	3.77	0.98	−0.60	−0.07	3.64	1.06	−0.58	−0.23
ER03	36. I remind myself of the potential negative influence of inappropriately using AI tools on academic integrity (e.g., originality, honesty, fairness, etc.)	3.74	0.99	−0.57	−0.03	3.87	1.02	−0.81	0.30

Abbreviations

AI	Artificial intelligence
AIC	Akaike's information criterion
ASH	Arts, humanities and social science
AVE	Average variance extracted
BR	Behavioral regulation
CFA	Confirmatory factor analysis
CFI	Comparative fit index
EA	Evaluation & assessment
EFA	Exploratory factor analysis
EFL	English as a Foreign Language
ER	Ethical regulation
HTMT	Heterotrait-monotrait ratio
LC	Language compensation
MNSQs	Mean squares
OMR	Output-focused metacognitive regulation
PP	Planning & performing
PR	Product refinement
RMSEA	Root mean square error of approximation
SRL	Self-regulated learning
SRMR	Standardized root mean square residual
STEM	Science, technology, engineering, and mathematics
TMR	Tool-focused metacognitive regulation

Supplementary Information

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Supplementary Material 1

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Author contributions

The first author designed the study, collected, analyzed and interpreted the data, and drafted and revised the manuscript; the second author collected the data, drafted part of and revised the manuscript; the third author collected part of the data and revised the manuscript.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests

The authors declare that they have no competing interests.

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